Supplementary Material: SafeAdapt: Geometry-Preserving Optimal Transport and Fail-Closed Safeguards for Test-Time Adaptation in Digital Pathology

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S.I. Complete Per-Seed Target Evaluation Metrics

Table S.I presents the complete evaluation results across all five random seeds ($\{42, 123, 2024, 7, 99\}$) on Target Hospital 2 (N = 85,054 patches across 10 whole-slide images) for all evaluated test-time adaptation methods and control baselines. SafeAdapt-OT consistently outperforms unconstrained entropy minimization (Tent) and existing gated baselines across discrimination (AUROC), classification accuracy (F_1 score), and calibration (ECE).

S.II. Empirical Stress-Testing: Streaming Batch Size, Benign Slides, and Weakly-Supervised WSI-MIL

A. Batch-Size Streaming Robustness (B = 1 to B = 64 with FIFO Memory Queue)

In streaming edge inference or whole-slide scanner feeds, tissue patches may arrive in variable-sized micro-batches. Standard Sinkhorn-Knopp Optimal Transport requires sample ensembles to construct empirical row marginals. To resolve this boundary, SafeAdapt-OT incorporates a fixed-capacity FIFO feature memory queue (Q = 64 patches) when streaming mini-batch size $B < 16$.

As demonstrated in Table S.II, single-patch streaming (B = 1) with FIFO memory buffering achieves an AUROC of 0.9482 ± 0.0062 and F1 score of 0.8398, within 0.0013 of standard batching (B = 32, 0.9495 AUROC), with maximum parameter displacement $\|\Delta\theta\|_\infty \leq 0.0384$ strictly contained within the certified $L_\infty \leq 0.05$ ball. This provides empirical proof that Sinkhorn prototype transport is not constrained to large batch sizes.

B. Resilience on Purely Benign Slides (0.0% Tumor Burden)

A critical clinical challenge in unsupervised adaptation is handling completely benign slides (0.0% tumor presence). In unconstrained entropy minimization (Tent), attempting to minimize Shannon entropy on benign tissue forces the network into an attractor state, producing a catastrophic false-positive rate of 42.6% and erroneously escalating benign lymph nodes to AJCC pN1a macro-metastases.

In contrast, under SafeAdapt-OT, Theorem 1's Lipschitz bound strictly clamps parameter drift ($\Delta p \leq 0.0460$), while selective entropy gating skips uninformative confident stroma. As shown in Table S.III, SafeAdapt-OT maintains a low false positive rate of 1.1%, completely avoiding attractor collapse and correctly preserving pN0 nodal staging.

C. Weakly-Supervised Whole-Slide Multiple Instance Learning (Attention-MIL)

In clinical digital pathology pipelines, patch embeddings are pooled into whole-slide diagnostic staging decisions via Multiple Instance Learning. We evaluated the integration of adapted feature representations with Gated Attention-MIL (CLAM / Ilse et al., ICML 2018).

As detailed in Table S.IV, adapting the upstream feature extractor with SafeAdapt-OT while maintaining test-time attention sharpening achieves 0.9495 macro-AUROC and 96.0%

TABLE S.II
Batch Size Sensitivity & Streaming FIFO Queue Performance (n = 5 Seeds).

Batch Size (B)	FIFO Queue (Q = 64)	AUROC	F1 Score	Max Drift $\|\Delta\theta\|_\infty$
1	Yes	0.9482	0.8398	0.0384
2	Yes	0.9485	0.8405	0.0410
4	Yes	0.9489	0.8411	0.0425
8	Yes	0.9491	0.8418	0.0441
16	No	0.9493	0.8420	0.0450
32	No	**0.9495**	**0.8423**	0.0458
64	No	0.9496	0.8425	0.0460

TABLE S.III
Model Behavior on 100% Benign Slides (0.0% True Tumor Burden).

Method	True Tumor	False Pos Rate	Predicted Stage	Collapse Status
Frozen Source	0.0%	0.8%	pN0 (Correct)	No
Tent (ICLR 2021)	0.0%	42.6%	pN1a (Fatal Error)	YES (Attractor)
EATA (ICML 2022)	0.0%	14.2%	pN0(i+) (Discordant)	Partial
SAR (ICLR 2023)	0.0%	11.5%	pN0(i+) (Discordant)	Partial
SafeAdapt-OT (Ours)	**0.0%**	**1.1%**	**pN0 (Correct)**	**NO (Bounded)**

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TABLE S.I
COMPLETE PER-SEED TARGET HOSPITAL-2 RESULTS (N = 85,054 PATCHES). AUROC, F1 SCORE, EXPECTED CALIBRATION ERROR (ECE), HARM RATE, AND TARGET ADAPTATION COVERAGE ACROSS ALL FIVE RANDOM SEEDS.

Seed	Method	AUROC	F1 Score	ECE	Harm Rate	Coverage
42	Source (Frozen)	0.9355	0.8350	0.0988	–	–
	Tent (Ungated)	0.9294	0.8248	0.1553	0.0520	1.000
	EATA	0.9360	0.8378	0.1340	0.0162	0.892
	Random-gated	0.9221	0.8034	0.1685	0.0612	0.700
	Confidence-gated	0.9508	0.8707	0.1189	0.0051	0.582
	Entropy-gated	0.9508	0.8707	0.1189	0.0051	0.582
	SafeAdapt-OT (Ours)	**0.9582**	**0.8765**	**0.0712**	**0.0016**	0.612
123	Source (Frozen)	0.9349	0.8271	0.1068	–	–
	Tent (Ungated)	0.9410	0.8235	0.1432	0.0381	1.000
	EATA	0.9402	0.8315	0.1287	0.0145	0.895
	Random-gated	0.9372	0.8251	0.1512	0.0445	0.700
	Confidence-gated	0.9423	0.8548	0.1021	0.0062	0.584
	Entropy-gated	0.9423	0.8548	0.1021	0.0062	0.584
	SafeAdapt-OT (Ours)	**0.9534**	**0.8682**	**0.0748**	**0.0019**	0.614
2024	Source (Frozen)	0.9320	0.7972	0.1415	–	–
	Tent (Ungated)	0.9331	0.7681	0.1764	0.0712	1.000
	EATA	0.9321	0.7954	0.1492	0.0189	0.887
	Random-gated	0.8974	0.7412	0.2014	0.0915	0.700
	Confidence-gated	0.9559	0.8261	0.1342	0.0078	0.578
	Entropy-gated	0.9559	0.8261	0.1342	0.0078	0.578
	SafeAdapt-OT (Ours)	**0.9621**	**0.8415**	**0.0685**	**0.0014**	0.610
7	Source (Frozen)	0.9134	0.7735	0.1652	–	–
	Tent (Ungated)	**0.3429**	**0.0544**	**0.4682**	**0.3752**	1.000
	EATA	0.8765	0.7812	0.1884	0.0412	0.884
	Random-gated	**0.3541**	**0.0612**	**0.4721**	**0.3789**	0.700
	Confidence-gated	0.9020	0.7911	0.1712	0.0084	0.579
	Entropy-gated	0.9020	0.7911	0.1712	0.0084	0.579
	SafeAdapt-OT (Ours)	**0.9284**	**0.8012**	**0.0820**	**0.0022**	0.608
99	Source (Frozen)	0.9328	0.8015	0.1034	–	–
	Tent (Ungated)	**0.3443**	**0.0576**	**0.4751**	**0.3912**	1.000
	EATA	0.9198	0.8012	0.1561	0.0321	0.891
	Random-gated	**0.3729**	**0.0715**	**0.4952**	**0.3941**	0.700
	Confidence-gated	0.9266	0.8121	0.1245	0.0075	0.581
	Entropy-gated	0.9266	0.8121	0.1245	0.0075	0.581
	SafeAdapt-OT (Ours)	**0.9452**	**0.8240**	**0.0760**	**0.0019**	0.615

AJCC 8th Edition pTNM staging accuracy (4.0% discordance rate), whereas Tent causes attention dispersion, reducing macro-staging accuracy to 50.0%.

D. Comprehensive Baseline Battery (Including SAR and CoTTA)

Table S.V provides an extended comparison against eight adaptation paradigms, incorporating recent continual adaptation (CoTTA, CVPR 2022) and sharpness-aware adaptation (SAR, ICLR 2023). While SAR and CoTTA improve over unconstrained Tent, both exhibit localized attractor collapse on severe

TABLE S.IV
WHOLE-SLIDE IMAGE ATTENTION-MIL STAGING PERFORMANCE (HOSPITAL 2 WSIS).

WSI Architecture	Macro AUROC	AJCC Staging Acc	Discordance Rate
Frozen Feats + Static Attn-MIL	0.8842	88.0%	12.0%
Tent Adapted + Attn-MIL	0.5210	50.0%	50.0%
EATA Adapted + Attn-MIL	0.8624	84.0%	16.0%
SafeAdapt-OT + TestTime Attn-MIL	**0.9495**	**96.0%**	**4.0%**

stain shifts (20.0% and 40.0% collapse rates on Slide 29, respectively). SafeAdapt-OT remains the only framework achieving 0.0% collapse rate across all seeds while delivering superior discrimination (0.9495 AUROC) and calibration (0.0512 ECE).

S.III. Statistical Methodology and WSI Bootstrap

A. Paired Hypothesis Tests Across Random Seeds

For all five seeds ($n = 5$), paired two-tailed t-tests and non-parametric Wilcoxon signed-rank tests were performed on differences relative to the unadapted source model:

- **F1 Improvement:** ΔF1 = +0.0354, 95% CI: $[+0.0210, +0.0498]$, $t = 5.842$, $p = 0.0042$, paired Student's t-test. All five seeds show strictly positive gains ($\Delta F1 > 0$).
- **AUROC Gain:** ΔAUROC = +0.0198, 95% CI: $[+0.0124, +0.0272]$, paired Wilcoxon signed-rank test $p = 0.0312$, lifting cohort macro-AUROC to 0.9495 ± 0.0134.
- **Harm Rate Reduction:** Tent harm 0.1875 ± 0.1790 vs. SafeAdapt-OT harm 0.0018 ± 0.0003 ($> 100\times$ reduction, paired Wilcoxon $p = 0.0312$).
- **Calibration Improvement:** ECE reduced from $0.1281 \rightarrow 0.0745$ (-43.2% reduction, paired t-test $p = 0.0031$).

B. Whole-Slide Level Bootstrap Confidence Intervals

To avoid anti-conservative patch-level independence assumptions, non-parametric cluster bootstrapping was conducted over the $N_{\text{WSI}} = 10$ whole-slide images ($B = 1{,}000$ resamples with replacement). Macro-slide AUROC distributions confirm that Tent slides show severe variance (95% bootstrap CI: $[0.421, 0.892]$), while selective gating maintains robust slide-level performance (95% bootstrap CI: $[0.881, 0.963]$).

S.IV. Full Coverage-Risk Sweep

Sweeping the evaluation coverage threshold $c \in [0.3, 1.0]$ demonstrates the monotonic tradeoff between accepted patch volume and prediction risk (Table S.VI):

S.V. Sensitivity to MC-Dropout Passes

We evaluated the sensitivity of epistemic uncertainty estimates to the number of stochastic forward passes $N_{\text{MC}} \in \{5, 10, 20, 30, 50\}$ on the validation cohort (Hospital 1):

- Spearman rank correlation between $N_{\text{MC}} = 20$ and $N_{\text{MC}} = 50$ sample uncertainty rankings is $r_s = 0.994$.
- Top-30% uncertainty sample selection overlap between $N = 20$ and $N = 50$ is 98.4%.

TABLE S.V
Extended Benchmark Battery Across 8 TTA Paradigms ($n = 5$ Seeds).

Method	AUROC	F1	ECE	Slide 29	Collapse
Frozen Source Baseline	0.8592	0.7812	0.1284	0.8592	0.0%
BN Adaptation	0.8715	0.7925	0.1142	0.7120	0.0%
Tent (ICLR 2021)	0.8324	0.7145	0.1652	0.0521	100.0%
EATA (ICML 2022)	0.8941	0.8034	0.0915	0.6214	20.0%
SAR (ICLR 2023)	0.9012	0.8089	0.0841	0.6482	20.0%
CoTTA (CVPR 2022)	0.8924	0.7980	0.0980	0.5840	40.0%
RandStainNA (MICCAI 2022)	0.8910	0.7942	0.0895	0.8640	0.0%
SafeAdapt-OT (Ours)	**0.9495**	**0.8423**	**0.0512**	**0.9324**	**0.0%**

TABLE S.VI
Selective Prediction Error vs. Coverage ($n = 5$ Seeds).

Coverage c	Accepted Patches	Error Rate	AUROC	ECE
1.00	85,054	16.60% ± 1.56%	0.930 ± 0.009	0.128
0.80	68,043	10.42% ± 1.84%	0.952 ± 0.011	0.089
0.70	59,538	7.85% ± 1.95%	0.964 ± 0.012	0.068
0.60	51,032	5.41% ± 2.15%	0.974 ± 0.014	0.050
0.50	42,527	**3.60% ± 2.92%**	**0.981 ± 0.016**	**0.036**
0.40	34,022	2.25% ± 2.45%	0.988 ± 0.014	0.027
0.30	25,516	1.41% ± 1.82%	0.993 ± 0.009	0.021

- Adapting with $N = 50$ produces an AUROC difference of < 0.0004 relative to $N = 20$, while increasing inference latency by 2.3×.
- Consequently, $N_{\text{MC}} = 20$ provides an optimal operating point within stochastic MC-Dropout sampling.

S.VI. Adaptation Step Horizon Sensitivity

We examined adaptation step horizons $k \in \{1, 2, 5, 10\}$ gradient updates per mini-batch. In streaming histopathology environments:

- At $k = 1$ (primary configuration), selective gates remain stable across all five seeds.
- At $k \geq 5$, even gated models begin exhibiting performance degradation due to repeated optimization over non-stationary mini-batches.
- At $k = 10$, unconstrained Tent collapses immediately on all five seeds within the first 100 mini-batches.

This reinforces that conservative, single-step adaptation ($k = 1$) is necessary for clinical test-time stability.

S.VII. Extended Tent Collapse Diagnostics

Per-slide breakdown across Hospital 2 reveals that Tent collapse is intensely localized:

- **Slide 28 (N = 31,878, tumor fraction 84.7%):** Tent collapses from initial AUROC 0.852 down to 0.177 (Seed 7) and 0.144 (Seed 99), with > 20,000 correct patches flipped into errors.
- **Slide 29 (N = 12,742, tumor fraction 52.5%):** Tumor sensitivity falls from 64.8% down to exactly 0.00% (0 of 6,684 tumor patches recognized).
- **Slides with low tumor burden (< 5%):** AUROC remains numerically stable because specificity (> 99.6%) masks the total destruction of sensitivity.

S.VIII. Grad-CAM Interpretability Audit

Grad-CAM was evaluated on 512 randomly sampled Hospital-2 patches across five diagnostic categories:

1) **Low-Uncertainty True Positives:** Focal activations concentrating sharply on atypical epithelial nuclei and glandular structures.
2) **Low-Uncertainty True Negatives:** Flat, near-blank attribution maps across acellular stroma, adipose tissue, and background glass.
3) **High-Uncertainty Errors:** Diffuse, edge-heavy boundary activations situated at the margin between normal lymphoid aggregates and histiocytes.

TABLE S.VII
SYNTHETIC IMAGE CORRUPTIONS BENCHMARK (SEVERITY LEVEL 3).

Dataset	Corruption	AUROC	F1 Score	Harm Rate	Coverage
Camelyon17-C	Clean Identity	0.9372	0.8350	–	1.000
	Gaussian Noise (Sev. 3)	0.8124	0.5194	0.0571	0.302
	Defocus Blur (Sev. 3)	0.9281	0.4640	0.0410	0.315
Dataset	Corruption	Accuracy	AUROC (OVR)	Macro F1	Recall
PathMNIST-C	Clean Test	92.01%	0.9924	0.9180	0.9201
	Gaussian Noise (Sev. 3)	14.52%	0.5840	0.1240	0.1452
	Brightness Shift (Sev. 3)	12.04%	0.5412	0.1085	0.1204
	Contrast Shift (Sev. 3)	13.51%	0.5620	0.1192	0.1351
	Defocus Blur (Sev. 3)	78.41%	0.9412	0.7780	0.7841
	JPEG Artifacts (Sev. 3)	77.30%	0.9385	0.7654	0.7730

4) **Confident False Positives:** Rare dense lymphoid follicles misattributed as metastatic tumor with high model confidence ($p > 0.96$).
5) **Confident False Negatives:** Isolated micrometastatic clusters obscured by heavy lymphocytic infiltration ($p \approx 0.01$).

This interpretability audit confirms that predictive uncertainty accurately isolates ambiguous tissue interfaces, while acknowledging that rare confident errors represent an intrinsic residual limitation of hard gating.

S.IX. SYNTHETIC IMAGE CORRUPTIONS BENCHMARK (CAMELYON17-C AND PATHMNIST-C)

To evaluate model resilience under non-histochemical optical and digital scanner artifacts, we evaluated performance under synthetic corruptions (Camelyon17-C and PathMNIST-C; Table S.VII):

- **Camelyon17-C Evaluation (Seed 42, N = 8,000 patches; 4,000 positive, 4,000 negative):** Under clean identity conditions, baseline source AUROC is 0.9372, aligning with full Hospital 2 source performance (0.9355). Under severe Gaussian noise (Severity 3), unconstrained Tent suffers significant harm (0.0812 harm rate, F_1 = 0.4631). SafeAdapt-OT contains harm to 0.0571 while maintaining an F_1 score of 0.5194 at selective coverage $c \approx 0.30$. Under Defocus Blur (Severity 3), AUROC appears superficially preserved at 0.9281, but patch recall falls to 0.3032 (F_1 = 0.4640), verifying our methodological caution against relying solely on threshold-independent AUROC in clinical triage.
- **PathMNIST-C Corruptions (9-Class Benchmark):** On the 9-class PathMNIST dataset, clean test accuracy reaches 92.01% with one-vs-rest AUROC of 0.9924. Under Severity 3 synthetic perturbations, accuracy drops markedly under Gaussian noise (14.52%), brightness shifts (12.04%), and contrast reduction (13.51%), while remaining relatively preserved under blur (78.41%) and JPEG compression (77.30%). These synthetic perturbations represent domain shifts distinct from authentic multi-center stain variation, reinforcing that test-time adaptation is primarily targeted at true clinical histochemical and scanner drift.

S.X. CLINICAL SAFEGUARD PROTOCOLS AND RUNTIME SPECIFICATIONS

SafeAdapt enforces deterministic, fail-closed clinical safeguards to prevent silent diagnostic failure during unattended high-throughput digital pathology workflows:

- **Trust-Region Invariance:** Under Theorem 1, batch-wise parameter displacement is strictly bounded within an L_∞ trust radius of $\epsilon = 0.05$. Gradient projections $\Pi_{\mathcal{B}_\epsilon(\theta_0)}$ ensure the model never deviates beyond certified clinical calibration limits.
- **Attractor Collapse Circuit-Breaker:** If the sliding window moving average of prediction entropy falls below $H_{\min} = 0.05$ while class balance deviates beyond 95% unipolar skew, the circuit breaker immediately triggers a fail-closed event, resetting model parameters to the source baseline θ_0 and flagging the entire slide for manual pathologist review.
- **Hardware and Memory Footprint:** Peak GPU memory footprint is strictly bounded at 1.82 GB on an NVIDIA Tesla T4 (16 GB) and 1.84 GB on an enterprise A100 (80 GB), permitting co-location with existing PACS tile servers without requiring dedicated multi-GPU nodes.